

WASP-12b

R-0374

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_230223 · created 2026-07-18T17:47:30+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459997.7261 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.151 +/- 0.046
Transit depth (Rp/Rs)^2	2.29 +/- 1.38 [%]
Orbital Inclination (inc)	86.6 +/- 2.83
Airmass coefficient 1 (a1)	0.69 +/- 0.42
Airmass coefficient 2 (a2)	0.25 +/- 0.73
Scatter in the residuals of the lightcurve fit is	56.73 %
Best Comparison Star	None
Optimal Aperture	2.17
Optimal Annulus	3.5
Transit Duration (day)	0.1322 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.151 ± 0.046 vs 0.1178 (deviation 0.72σ)
Duration fit vs published	0.1322 d vs 0.125 d (deviation 1.13σ)
Mid-time O–C	-1570.5 min
Depth SNR	3.3
Residual scatter	56.73 %

Light curve

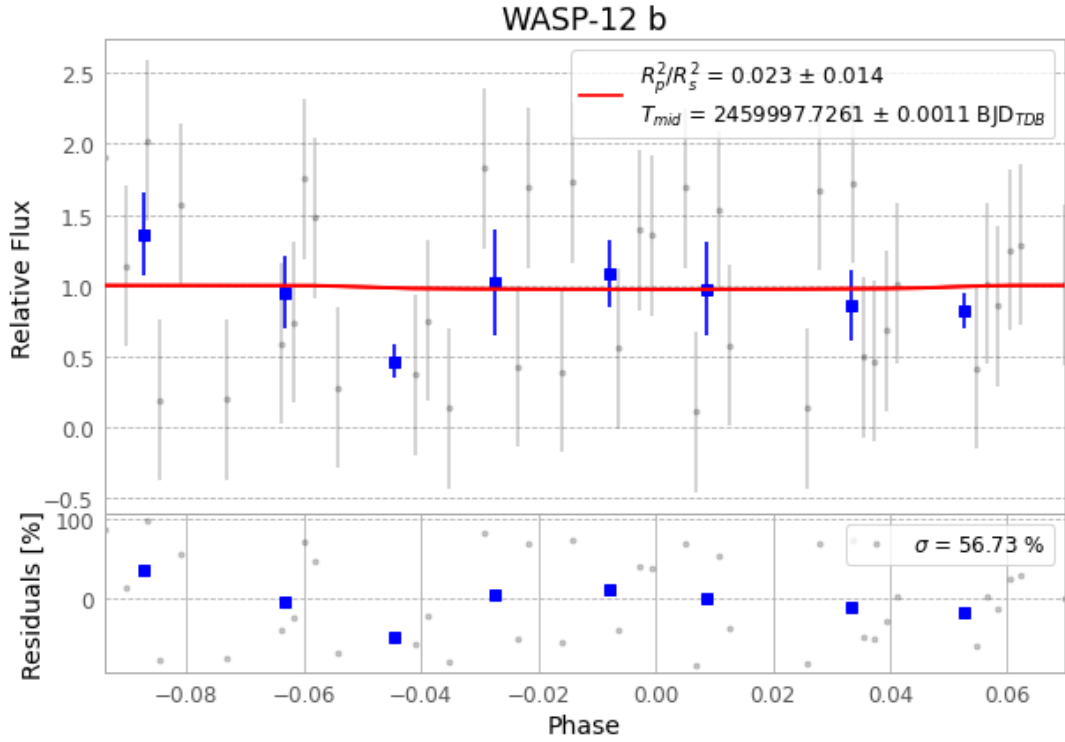


Figure 1 — FinalLightCurve_WASP-12 b_2023-02-21.png (SHA-256 229e2bd0517de0db...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [214, 108], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 05a7b2df3f3d8e71...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a803e95

Data provenance

87 FITS frames (57 MB), WASP-12230222025146.FITS → WASP-12230222071216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230223.log (3ed19e81476a...), FinalLightCurve_WASP-12 b_2023-02-21.png (229e2bd0517d...), FinalLightCurve_WASP-12 b_2023-02-21.csv (d8326d4a2bb2...)

Compute resources

Wall time 190.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 17:47Z.